AN ALGEBRAIC APPROACH TO NECESSITIES

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Abstract. Bacon (2018, 2024) gives an account of necessities in higher-order logic by internalizing the *proof theory* of normal modal logics. I give an alternative account that internalizes their algebraic semantics instead: a necessity is a property of propositions that maps the tautology to itself and commutes with conjunction. To properly formalize this definition we need to get clearer on exactly what sort of structure propositions form under entailment. I introduce *intensional* counterparts of familiar order-theoretic and algebraic concepts and model algebraic necessities as operations that interact with intensional algebraic structure the right way. I show that Bacon's definition and mine are equivalent, then apply my algebraic characterization to study the intensional algebraic structure that necessities themselves form. I prove that the intensional lattice of necessities is distributive (solving an open problem from Bacon and Zeng 2022), and in fact has the structure of an (intensional) Heyting algebra. I study pseudocomplements of necessities in this structure, which correspond on Kripke frames to modalities defined via *in*accessibility relations. Classical principles on pseudocomplementation correspond to philosophically interesting properties of necessities. Excluded middle corresponds to the idea that a necessity is *generated* by a single proposition (necessarily admits a strongest necessary proposition); double negation elimination corresponds to the idea of an *infinitely closed* necessity.

Introduction

The metaphysics of modality has traditionally focused on the study of particular notions of necessity, such as metaphysical necessity, nomic necessity, historical necessity, and so on (see, e.g., Fine, 2002). Recently, however, there has been a shift towards the systematic study of *necessities* as a whole. A big driver of this shift is that some questions about particular notions of necessity are best answered by appealing to the general structure of necessities. Most prominently, Kripke (1980, p. 99) introduces the notion of metaphysical necessity as "necessity in the highest degree—whatever that means." Systematic theories of necessities have been developed for the explicit purpose of making sense of this idea, by giving precise accounts of what it means for a necessity to be of higher degree or *broader* than another, and what sort of structure necessities form under this relation (Hale, 2013; Williamson, 2016; Bacon, 2018; Roberts, 2025).

There are also many general questions about necessities that are of independent interest, and can be fruitfully studied in abstraction from any particular notion of necessity. For example, what does it mean for a necessity to be a restriction of another (Smiley, 1963; Humberstone, 1981; Hale and Leech, 2017; Roberts, 2020; Bacon and Zeng, 2022, Sec. 3.4)? Can all intensional notions be reduced to extensional ones (Lewis, 1986; Bacon, 2018, 2024, Ch. 7)? More generally, clarifying the structure of necessities is interesting in its own right. The concept of a necessity is

central to metaphysics—no less central than, say, those of parthood and location—and metaphysics, like any other part of philosophy, should seek as complete an account as possible of the structure of its central concepts.

The theory of necessities developed in Bacon (2024, Ch. 7) is an ideal environment to study questions of this sort. Bacon gives a reductive analysis of what a necessity is and what it means for a necessity to be broader than another in purely logical terms, meaning that many questions about the structure of necessities can be answered through logical means alone. The general idea driving Bacon's approach is that necessities are to the world what normal modal operators—the boxes of modal logic—are to language. Bacon makes this intuition precise by internalizing the *proof theory* of normal modal logics. Syntactically, a normal modal logic for an operator M is a set of sentences that (i) contains all tautologies, (ii) contains every instance of the K-axiom for M, and is closed under the rules of (iii) modus ponens and (iv) necessitation for M. Mirroring this definition, Bacon defines in the object language the property of being a *normal theory* for a property of propositions M. He then defines a *weak necessity* to be a property of propositions whose every normal theory is true, and a *necessity* proper as something that is necessarily a weak necessity, according to every weak necessity.

It is natural to consider a different way of implementing the same general idea: instead of internalizing the proof theory of normal modal logics, we can internalize their *semantics*. In this article, I develop an approach along these lines that internalizes the *algebraic semantics* of normal modal logics. Algebraically, propositional modal languages are interpreted over *modal algebras* (Jónsson and Tarski, 1951): Boolean algebras equipped with an operator—used to interpret the box—that maps the top element to itself and commutes with finite meets. These equational conditions can be straightforwardly internalized in the object language. A necessity, on my account, is simply a property of propositions that maps the tautologous proposition to itself and commutes with conjunction.

Carrying out this program requires some care. Modal algebras, like Boolean algebras, lattices, and partial orders, are defined in terms of *extensional* notions: they concern how things in fact stand with the elements of a given set. But propositions do not merely happen to stand in the relations characteristic of a Boolean algebra. They do so as a matter of broad necessity, and they would continue to do so even if "new" propositions came into existence. To internalize the algebraic semantics faithfully, then, we need *intensional* counterparts of the familiar extensional notions of order theory and algebra. I develop such counterparts in Section 3, using purely equational formulations that do not presuppose the notion of broad necessity.

With these intensional notions in hand, I show that the algebraic characterization of necessities is equivalent to Bacon's original proof-theoretic one: to be a necessity in the algebraic sense just is to be a necessity in Bacon's sense. I then apply my algebraic characterization of necessities to prove general results about the *algebra of necessity*—the algebraic structure that the family of necessities forms under the entailment relation. Bacon and Zeng (2022) proved that necessities in Bacon's sense form an (intensional) bounded lattice with respect to entailment, but left open the question of whether the lattice is distributive. My algebraic characterization of necessities affords a simple proof that it is. What is more, I prove that necessities satisfy an infinite distributivity law that gives them the structure of an *intensional*

Heyting algebra, the intensional analog of the sort of algebraic structure typically used to model the intuitionistic propositional calculus.

With Heyting algebra structure at hand, we can make sense of the operation of taking the *pseudocomplement* of a necessity. If we think of an M-necessary proposition as one that is true at all M-possible worlds, then the pseudocomplement of a necessity M is the property of being true at all M-*impossible* worlds (Humberstone, 1983; Gargov et al., 1987); the pseudocomplement operation is a generalization of this idea that also applies to necessities that cannot be characterized in terms of possible worlds this way.

The pseudocomplement operation does not in general behave classically. I show how to characterize different intuitive senses in which a necessity can be *defective* in terms of which classical principles—like excluded middle and double negation elimination—it fails to satisfy. The property of propositions *having probability 1*, for example, is arguably a necessity in the operative sense, but is defective in at least two ways. First, it is not *principal*: for every proposition that has probability 1, we can find a strictly stronger proposition that also has probability 1; there is no maximally strong proposition with probability 1. Second, it is not *infinitely closed*: it fails to make necessary every logical consequence of the propositions it makes necessary (Bacon and Zeng, 2022; Bacon, 2024, Ch. 7). Many paradigmatic examples of necessity, like metaphysical necessity and nomic necessity, do not seem to be defective in either way. I give a general characterization of pseudocomplements and use it to show that a necessity is principal iff its join with its pseudocomplement is the narrowest necessity (excluded middle), and that a necessity is infinitely closed iff it is equal to its double pseudocomplement (double negation elimination).

The article is organized as follows. Section 1 briefly introduces Classicism. Section 2 reviews Bacon's proof-theoretic theory of necessities. Section 3 develops intensional order-theoretic and algebraic concepts. Section 4 uses these notions to define algebraic necessities and proves that they coincide with necessities in Bacon's sense. Section 5 studies the intensional lattice of necessities. Section 6 studies pseudocomplements of necessities.

1. Preliminaries

We work in a standard simply typed higher-order language. The *types* of our language consist of the basic types e, t of singular terms and sentences respectively, and the functional type $\sigma \to \tau$ for any types σ, τ. We assume there is a countably infinite set of variables Var^σ for each type σ. A *signature* is a typed family of sets $\Sigma^\cdot$ with Σ^σ a set of constant symbols for each type σ. The *terms* of our language are defined as those we can recursively build from variables and constants via application and λ-abstraction, in the usual way. We write $A : \sigma$ to indicate that A is a term of type σ. Throughout, we work in an unspecified signature containing the usual Boolean connectives and quantifiers at every type. Some useful abbreviations are collected in Table 1.

Our background higher-order logic is `H`, a standard classical higher-order logic in which $\beta\eta$-equivalence implies identity. A full axiomatization can be found in Bacon (2024, Ch. 5). We will work in an extension of `H` introduced in Bacon and Dorr (2024) called *Classicism* (`C`), obtained by closing the standard classical higher-order logic `H` under the schematic rule:

$$\text{if } \vdash P \leftrightarrow Q, \text{ then } \vdash \lambda\bar{x}.P = \lambda\bar{x}.Q. \tag{Equiv}$$

$$
\begin{array}{ll}
\forall \bar{x} M := \forall x_0 \cdots \forall x_n M & \lambda \bar{x}.M := \lambda x_0 \ldots x_n.M \\
\heartsuit_t := \heartsuit \quad (\heartsuit \in \{\wedge, \vee, \rightarrow, \leftrightarrow\}) & \heartsuit_{\sigma\rightarrow\tau} := \lambda X Y z^{\sigma}.X z \heartsuit_{\tau} Y z \\
\top := \exists^{(e\rightarrow t)\rightarrow t\rightarrow t} \exists^{(e\rightarrow t)\rightarrow t} & \bot := \neg\top \\
\bot_t := \bot & \bot_{\sigma\rightarrow\tau} := \lambda x^{\sigma}.\bot_{\tau} \\
\top_t := \top & \top_{\sigma\rightarrow\tau} := \lambda x^{\sigma}.\top_{\tau} \\
=_{\sigma} := \lambda X Y. \forall Z^{\sigma\rightarrow t}(ZX \leftrightarrow ZY) & \subseteq_{\sigma} := \lambda X Y^{\sigma\rightarrow t}. \forall Z(XZ \rightarrow YZ) \\
F[\bar{A}] := FA_0 \wedge \cdots \wedge FA_n &
\end{array}
$$

TABLE 1. Abbreviations. Here $\bar{x} = (x_0, \ldots, x_n)$ is any sequence of variables and $\bar{A} = (A_0, \ldots, A_n)$ is any sequence of terms of the same type.

This rule ensures that provably coextensive properties are identical. In particular, it ensures that there is exactly one logical truth, $\top$, as any two logical truths (theorems of H) are provably equivalent in H.

In Classicism, we can give an algebraic definition of *entailment*, both for propositions and for higher-order relations. We let

$$\leq_{\sigma} := \lambda X Y. X \wedge_{\sigma} Y = X.$$

By (Equiv), a sufficient condition for X entailing Y is that we can prove that X materially implies Y, since $X\bar{x} \rightarrow Y\bar{x}$ is equivalent to $X\bar{x} \wedge Y\bar{x} \leftrightarrow X\bar{x}$.

It will be helpful to work with a seemingly more general version of (Equiv), the *conditional equivalence* rule, which actually turns out to also be a derived rule of Classicism.

$$\text{if } \vdash P = \top \rightarrow (Q \leftrightarrow Q'), \text{ then } \vdash P = \top \rightarrow \lambda \bar{x}.Q = \lambda \bar{x}.Q'. \qquad \text{(CEquiv)}$$

Here we must also impose the requirement that $\bar{x} \notin \mathsf{FV}(P)$. This rule gives us a simple way to establish identities while reasoning under a hypothesis. We can do that, for example, whenever the hypothesis is a proposition that entails its own identity with $\top$. I will call a proposition with this property *persistent*:

$$\mathsf{Persistent} := \lambda p.p \leq (p = \top).$$

When P is persistent and we can derive $(Q \leftrightarrow Q')$ from P, we can use (CEquiv) to conclude that $\lambda \bar{x}.Q = \lambda \bar{x}.Q'$ on the assumption of P. For $P = \top$ entails P, so if we can derive $(Q \leftrightarrow Q')$ from P we can also derive it from $P = \top$. This means we can use (CEquiv) to show we can derive $\lambda \bar{x}.Q = \lambda \bar{x}.Q'$ from $P = \top$, and if P is persistent then we can also derive it from P itself.

2. BACON'S THEORY OF NECESSITIES

Bacon (2024, Sec. 7.2) introduces his logical notion of necessity in Classicism. As noted, the guiding idea is to internalize the proof theory of normal modal logics into the object language. To do that, we first define what it is for a property of propositions to be a *normal theory* for a property of propositions. A normal theory is a worldly analog of the notion of a normal modal logic: a set of sentences

containing all tautologies, every instance of the K-axiom, and closed under modus ponens and necessitation. We begin by defining:

$$\mathsf{Taut} := \lambda X.X\top \qquad \mathsf{MP} := \lambda X.\forall pq(Xp \wedge X(p \to q) \to Xq)$$
$$\mathsf{NecRule} := \lambda XY.\forall p(Xp \to X(Yp)) \qquad \mathsf{K} := \lambda XY.X(\mathsf{MP}(Y))$$

We then put:

$$\mathsf{NTh} := \lambda X^{t\to t}Y^{t\to t}.\mathsf{Taut}X \wedge \mathsf{MP}X \wedge \mathsf{NecRule}XY \wedge \mathsf{K}XY$$

Intuitively, $\mathsf{NTh}XY$ says that X is a normal theory for the operator Y. That is, X applies to the tautologous proposition, is closed under modus ponens and necessitation for Y, and says that Y itself is closed under modus ponens, which mirrors the K-axiom.

Definition 2.1. A *weak necessity* is defined as a property of propositions such that whatever belongs to all of its normal theories is true:

$$\mathsf{WNec} := \lambda X.\forall p(\forall Y(\mathsf{NTh}YX \to Yp) \to p)$$

A *necessity* is a property of propositions that is necessarily a weak necessity, relative to every weak necessity:

$$\mathsf{Nec} := \lambda X.\forall Y(\mathsf{WNec}Y \to Y(\mathsf{WNec}X)).$$

Bacon defines a necessity M to be *broader* than a necessity N if M entails N, *narrower* if M is entailed by N. As is now well known, Classicism proves that there is a *broadest* necessity, $\Box$. It can be defined as the property of having every necessity:

$$\Box := \lambda p.\forall X(\mathsf{Nec}X \to Xp),$$

and shown to be equivalent to the property of being identical to the tautologous proposition,

$$\mathsf{C} \vdash \Box = (\lambda p.p = \top).$$

In other words, the broadest necessity is that which makes only tautologies necessary.

We can use the broadest necessity to characterize entailment as *broadly strict implication*:

$$\mathsf{C} \vdash \leq_\sigma = \lambda XY^\sigma.\Box\forall \bar{z}(X\bar{z} \to Y\bar{z}),$$

where σ is any relational type. This makes clear that Classicism comes with a *coarse-grained* metaphysical picture: broadly necessarily equivalent properties and propositions are identified.

3. From Extensional to Intensional

My account of necessities describes a necessity as a property of propositions that interacts the right way with the algebraic structure that propositions form under entailment. To properly formulate it, we need to get clear on what, exactly, this structure consists in. This section describes it, and Section 3.2 motivates the resulting intensional notions by appeal to the category of modalized sets.

3.1. **Intensional Lattice Theory.** It is often said that, in Classicism, propositions form a Boolean algebra with respect to entailment. This is true, but it is only a partial description. That of a Boolean algebra is an *extensional* concept. A Boolean algebra is a special kind of poset. A poset is a set equipped with a reflexive, antisymmetric, and transitive binary relation. Those of reflexivity, antisymmetry, and transitivity are all extensional notions. In a higher-order setting, they are most naturally formalized as follows:

$$\begin{aligned}\mathsf{Refl} &:= \lambda R.\forall x\, Rxx\\ \mathsf{Antisymm} &:= \lambda R.\forall xy((Rxy \wedge Ryx) \to x = y)\\ \mathsf{Trans} &:= \lambda R.\forall xyz((Rxy \wedge Ryz) \to Rxz).\end{aligned}$$

Propositional entailment satisfies all three conditions. So does the restriction of the *is as large as* relation to planets in our solar system. But there are important differences in how modally robustly these two relations satisfy the three conditions. As it turns out, no two planets in our solar system are exactly the same size. But they could have been (in some sense of 'could'), in which case the *is as large as* relation restricted to planets in our solar system would have failed to be antisymmetric. Moreover, the Earth is in fact larger than Mercury, but things could (in some sense of 'could') have been the other way around. The position that planets occupy in the structure determined by the *is as large as* relation is merely contingent.

Entailment is different in both respects. Entailment is *broadly necessarily* a partial order, in every sense of necessity. And if p entails q, then this holds of broad necessity. This claim, too, holds of broad necessity. In particular, had there been "new" propositions r and s such that r entails s, this would also hold as a matter of broad necessity.

These intensional order-theoretic features of entailment can be precisely formulated without presupposing Bacon's notion of broad necessity, using higher-order identities instead. The properties of being broadly necessarily reflexive, antisymmetric, and transitive can be equivalently formulated as follows:

$$\begin{aligned}\mathsf{Refl!} &:= \lambda R.(\lambda x.Rxx) = (\lambda x.\top)\\ \mathsf{Antisymm!} &:= \lambda R.(\lambda xy.(Rxy \wedge Ryx)) \leq =\\ \mathsf{Trans!} &:= \lambda R.(\lambda xyz.(Rxy \wedge Ryz)) \leq (\lambda xyz.Rxz)\end{aligned}$$

To bear R to oneself just is to be such that $\top$. Bearing R to one another entails being equal; being connected by R in two steps entails being connected by R in one step. We call these notions *intensional* reflexivity, antisymmetry, and transitivity.

The second feature—that, broadly necessarily, a proposition that entails another does so as a matter of broad necessity—can be captured through the notion of a *persistent relation* (generalizing the notion of a persistent proposition from Section 1). Intuitively, a relation is persistent when, broadly necessarily, if some things have it then they have it as a matter of broad necessity. We can give an equational formulation of this idea as well:

$$\mathsf{Persistent} := \lambda R.R \leq (\lambda \bar{x}.R\bar{x} = \top)$$

We will call a relation an *intensional partial order* when it is persistent and intensionally reflexive, antisymmetric, and transitive:

$$\mathsf{PO!} := \lambda R.\mathsf{Persistent}R \wedge \mathsf{Refl!}R \wedge \mathsf{Antisymm!}R \wedge \mathsf{Trans!}R.$$

It is straightforward to verify that, in Classicism, entailment is an intensional partial order, at any type.

There is a similar dialectic involving the other components in the definition of a Boolean algebra. A Boolean algebra is a *lattice*, meaning a partial order where any two elements have both a meet and a join. The notions of meet and join are extensional notions. Focusing specifically on meets and joins with respect to entailment, their most natural higher-order regimentation is as follows:

Definition 3.1 (Bounds)**.** A *lower bound of* $Y : \sigma \to t$ *in* Z is something to which Z applies that entails anything that falls under Y, and an *upper bound* of $Y : \sigma \to t$ *in* Z is something to which Z applies that is entailed by anything that falls under Y:

$$\mathsf{LB} := \lambda x Y Z.Zx \wedge \forall w(Yw \to x \leq w),$$
$$\mathsf{UB} := \lambda x Y Z.Zx \wedge \forall w(Yw \to w \leq x).$$

An *extensional meet of* Y *in* $Z : \sigma \to t$ is a lower bound of Y in Z that is entailed by every other lower bound of Y in Z, and an *extensional join* of Y *in* $Z : \sigma \to t$ is an upper bound of Y in Z that entails every other upper bound of Y in Z:

$$\mathsf{Meet} := \lambda x Y Z.\mathsf{LB}xYZ \wedge \forall w(\mathsf{LB}wYZ \to w \leq x),$$
$$\mathsf{Join} := \lambda x Y Z.\mathsf{UB}xYZ \wedge \forall w(\mathsf{UB}wYZ \to x \leq w).$$

I adopt the convention of writing $\mathsf{LB}xY$—pronounced *x is a lower bound of Y*—for $\mathsf{LB}xY\top_{\sigma \to t}$ and similarly for the other cases. A lower (upper) bound of a pair of entities is defined as a lower (upper) bound of the property of being identical to one or the other, and likewise for extensional meets and joins. In these cases we may write $\mathsf{LB}x[y, z]Z$ for $\mathsf{LB}x\lambda v.v = y \vee v = zZ$, and similarly for the other cases.

It is easy to see that conjunction and disjunction respectively give extensional meets and joins of pairs of entities (at any relational type) with respect to entailment. But again, this is not the whole story. Entities at any relational type have meets and joins with respect to entailment as a matter of broad necessity, and these meets and joins are necessarily given by their conjunctions and disjunctions. To capture this, we can introduce *intensional* notions of bounds, which again we can do in a purely equational way without assuming the notion of broad necessity.[1]

Definition 3.2 (Intensional Bounds)**.** An *intensional lower bound* of $Y : \sigma \to t$ in Z is some x with Zx such that Y entails the property of being entailed by x, and an *intensional upper bound* of $Y : \sigma \to t$ in Z is some x with Zx such that Y entails the property of entailing x:

$$\mathsf{LB!} := \lambda x Y Z.Zx \wedge Y \leq (\lambda w.x \leq w) \qquad \mathsf{UB!} := \lambda x Y Z.Zx \wedge Y \leq (\lambda w.w \leq x).$$

An *intensional meet* of Y in Z is an intensional lower bound x of Y in Z such that being an intensional lower bound of Y in Z entails the property of entailing x, and an *intensional join* of Y in Z is an intensional upper bound x of Y in Z such that being an intensional upper bound of Y in Z entails the property of being entailed

[1] Intensional meets (their 'modalized greatest lower bounds') are introduced by Bacon and Zeng (2022, Sec. 5), who argue that they are the appropriate notions to use in non-extensional contexts drawing from category theory. I expand on their argument towards the end of this section.

by x:

$$\mathsf{Meet!} := \lambda xYZ.\mathsf{LB!}xYZ \wedge (\lambda w.\mathsf{LB!}wYZ) \leq (\lambda w.w \leq x),$$
$$\mathsf{Join!} := \lambda xYZ.\mathsf{UB!}xYZ \wedge (\lambda w.\mathsf{UB!}wYZ) \leq (\lambda w.x \leq w).$$

I use similar notational conventions for intensional bounds as for extensional bounds.

It is important to understand how the extensional notions relate to their intensional counterparts. Clearly, if x is an intensional lower bound of Y in Z, then it is also an extensional lower bound of Y in Z, and the same holds for upper bounds. The converse is false: something could *in fact* entail every Y but not entail merely possible instances of Y.

Meets and joins are more complicated. An extensional meet of Y in Z need not be an intensional meet of Y in Z. Say that a relation is *rigid* when, so to speak, it can neither lose nor gain instances across modal space. Formally, a relation is rigid when:

$$\mathsf{Rigid} := \lambda R.(\lambda S.R \leq S) = (\lambda S.R \subseteq (\lambda \bar{x}.S\bar{x} = \top))$$

Suppose we add *Rigid Comprehension* (Dorr et al., 2021; Goodsell, 2022; Bacon and Dorr, 2024), the principle that every property is coextensive with a rigid property, to Classicism. Let X be a persistent but non-rigid property of propositions, and let X' be a rigid property coextensive with X. Then the proposition that every X' is true, i.e., $\forall p(X'p \to p)$, is an extensional meet of X, but not an intensional meet. Because X is persistent and the actual Xs are exactly the X's, $\forall p(X'p \to p)$ entails every X and is entailed by anything that entails every X. However, were there any "new" Xs, $\forall p(X'p \to p)$ might have failed to entail them. So, X need not entail the property of being entailed by $\forall p(X'p \to p)$, in which case $\forall p(X'p \to p)$ is not an intensional meet of X. The intensional meet of X is instead the proposition that every X is true, i.e., $\forall p(Xp \to p)$. If, however, we modify the example so that X is rigid, then $\forall p(Xp \to p)$ is both an extensional and an intensional meet of X.

Perhaps surprisingly, an intensional meet of Y in Z need not be an extensional meet of Y in Z, either. There are models of Classicism where not every persistent property of propositions has an extensional meet (or join).[2] On the other hand, we can prove in Classicism that every persistent property of propositions Y has both an intensional meet and an intensional join, given respectively by the propositions $\forall p(Yp \to p)$ and $\exists p(Yp \wedge p)$. Moreover, as can be inferred from the argument in the previous paragraph, the intensional meet and join of a rigid property are also its extensional meet and join. This is true at any relational type, not just of properties of propositions.

A consequence of this last claim is that *binary* extensional meets and joins always coincide with their intensional counterparts, since the property of being identical to either of two given entities is always rigid. Accordingly, I will henceforth speak simply of meets and joins in finitary contexts.

Finally, we can introduce the notion of an *intensional lattice* with respect to entailment. It is straightforward to verify that, in Classicism, all the notions of intensional bounds just introduced are persistent relations. So, we can say that an intensional lattice consists of a persistent property where, as a matter of broad necessity, any two entities have both an intensional meet and an intensional join. Persistence will guarantee that intensional meets and joins do not change across modal space. This can be captured equationally as follows:

[2] I give one in Appendix A; see also Bacon and Dorr (2024, Prop. D5).

Definition 3.3 (Intensional Lattice)**.** An *intensional lattice* with respect to entailment is a persistent property X such that the relation two things bear when they are both instances of X is the same as the relation they bear when they have an intensional meet (resp. join) in X:

$$\mathsf{Lat!} := \lambda X.\mathsf{Persistent}X \wedge (\lambda yz.X[y,z]) = (\lambda yz.\exists w\, \mathsf{Meet!}w[y,z]X) \\ \wedge (\lambda yz.X[y,z]) = (\lambda yz.\exists w\, \mathsf{Join!}w[y,z]X).$$

We can do similar things with the remaining components of the definition of a Boolean algebra, defining the notions of *being intensionally distributive* (see Section 5.3), *being intensionally bounded* and *having intensional complements*. This ultimately leads to the notion of an *intensional Boolean algebra*. Propositions form an intensional Boolean algebra with respect to entailment.

3.2. **A Connection to Modalized Sets.** Building on some remarks by Bacon and Zeng (2022, Sec. 5 and n. 74), I want to suggest that the intensional lattice-theoretic notions just introduced are very natural in the context of Classicism. They are what the familiar lattice-theoretic definitions become when those definitions are generalized to the category of *modalized sets*, in which Classicism is perspicuously modeled.

Categorically, a *modalized set* is a functor D from a preorder $(W, \preceq)$, viewed as a thin category,[3] to the category **Set** of sets.[4] When f is the unique arrow from w to v, the image D_f is a function from D_w to D_v that describes which elements of D_v represent which elements of D_w. We write this function as $i_{wv} : D_w \to D_v$. Since D is a functor it must preserve the identity arrow and composition, meaning i_{ww} is the identity function on D_w and $i_{uv} \circ i_{wu} = i_{wv}$ whenever $w \preceq u \preceq v$.

A modalized set can be used to model propositions in the following way. We start with a rooted preorder $(W, \preceq, @)$ representing possible worlds with an accessibility relation. To define the functor D, we let D_w equal the powerset of the set of worlds that w "sees," and whenever $U \in D_w$ we let $i_{wv}(U)$ be the *truncation* of U about v, i.e., $i_{wv}(U) = U \cap \{v' \in W : v \preceq v'\}$. The intuitive idea is that each element of $D_@$ and its images under the maps $i_{@v}$ for $v \succeq @$ represent one proposition. The subset $i_{@v}(U)$ is the "proposition stage" representing the same proposition at v that U represents at $@$.

Intensional meets and joins are the natural notions of meet and join in the category of modalized sets in the sense that they are the notions you get when you formulate the familiar definitions of meets and joins in what is called the *internal logic* of the category of modalized sets. Here is the idea. In a poset $(A, \leq)$, an infinitary meet operation is an operation $\bigwedge : \wp(A) \to A$ satisfying the usual condition:

$$a \leq \bigwedge S \iff a \leq s \text{ for every } s \text{ in } S. \qquad (\star)$$

The notion of a powerset is an instance of the more general category-theoretical concept of a *power object*, which internalizes the idea of a collection of *subobjects* within a category. A subobject of an object X in a category is an equivalence class

[3] This means that the objects are the elements of W, and exactly one arrow from w to v exists whenever $w \preceq v$.

[4] In other words, modalized sets are the objects of the functor category $\mathbf{Set}^W$, a presheaf topos, whose internal logic is given by Kripke–Joyal semantics and is in general intuitionistic; see Mitchell and Moggi (1991) and Mac Lane and Moerdijk (1992, Chs. I, VI.6–7).

of monomorphisms into X, capturing the idea of a subset in the categorical setting. The power object of X then represents the collection of all such subobjects, generalizing the notion of a powerset from set theory to an arbitrary category. In the category of modalized sets, subobjects correspond not to subsets, but to *subdomains*. A subdomain of a modalized set D is another modalized set E based on the same underlying preorder $(W, \preceq)$, such that $E_w \subseteq D_w$ for every $w \in W$ and where the counterpart function agrees with that of D whenever both are defined: $i^D_{wv}a = i^E_{wv}a$ whenever $a \in E_w$. In a modalized set thought to model propositions, these correspond one to one with the persistent properties of propositions: a subdomain of a modalized set D that represents propositions chooses a collection of proposition stages at each world and identifies representatives thereof at each subsequent world.

The other thing we need to do to read ($\star$) in the internal logic of modalized sets is to interpret the quantifiers. The internal logic of a category must interpret expressions in a way that makes them natural transformations. In the case of propositional quantification, this means we want to find an operation from persistent properties of propositions to propositions that commutes the right way with the counterpart functions. This naturality condition holds if we interpret quantification in a modalized set D representing propositions as the mapping Π.

$$\Pi_w(F) := \{v \succeq w : F_u = D_u \text{ for all } u \succeq v\},$$

where F is a subdomain of D restricted to the worlds above w. Informally, $\Pi_w(F)$ consists of all worlds v above w where every proposition is F, and this remains the case at all successors.

If, in ($\star$), we take S to range over subdomains and interpret the quantifier 'for every' as just spelled out, we get exactly the notion of intensional meet. The reason why the object language definition of intensional meet looks more complicated is that, while Classicism has natural models based on modalized sets (Bacon, 2024, Secs. 17.1 and 18.1), the underlying higher-order logic is not the internal logic of modalized sets, because quantifiers are not interpreted as natural transformations in the model theory, and the connectives are the Boolean ones rather than the internal (intuitionistic) ones. But we can express the notion of quantification that is internal to modalized sets by necessitating the quantifiers, which is precisely how we get the definition of intensional meet from its extensional counterpart.

4. Algebraic Necessities

Now that I have clarified what kind of algebraic structure on propositions we should pay attention to, I proceed to internalize the algebraic semantics of modal logics with respect to that kind of algebraic structure to define the notion of an algebraic necessity.

The central concept of the algebraic semantics of normal modal logics is that of a *modal algebra* (e.g., Chagrov and Zakharyaschev, 1997, Ch. 7). A modal algebra is simply a Boolean algebra equipped with an operator that maps the top element to itself and commutes with finite meets. This is a purely equational definition that can be internalized in the object language. The idea here is that propositions form a Boolean algebra structure with respect to entailment, where conjunction and negation give meets and complements respectively. An *algebraic necessity*, then, is just a property of propositions that satisfies appropriate generalizations of

the defining equations of modal algebras with respect to the underlying Boolean algebra of propositions—or rather, as we will see, with respect to the *intensional* Boolean algebra structure of propositions in the sense of Section 3.

In Section 4.1 I give a precise definition of algebraic necessities, drawing on the intensional notions of Section 3, and show that they are precisely the necessities in Bacon's sense. In Section 4.2 I give a further characterization of necessities of a similarly algebraic flavor, in terms of an intensional notion of *filter*.

4.1. **Algebraic Necessities Defined.** When we think about internalizing the defining conditions of modal algebras, we want to make sure our resulting notion of an algebraic necessity is well behaved with respect to the underlying intensional Boolean algebra structure of propositions, not merely the extensional portion thereof. This motivates defining algebraic necessities as follows:

Definition 4.1 (Algebraic Necessity)**.** A property of propositions M is an *algebraic necessity* if it maps the tautologous proposition to itself, and the relation two propositions bear when their conjunction falls under M is the same as the relation they bear when both fall under M:

$$\mathsf{AlgNec} := \lambda M.(M\top = \top) \wedge (\lambda pq.M(p \wedge q)) = (\lambda pq.Mp \wedge Mq)$$

The first conjunct is a straightforward regimentation of the condition on modal algebras that the modal operator maps the top element to itself. An immediate corollary of the broad necessity of identity is that an algebraic necessity necessarily maps the tautologous proposition to itself, in every sense of necessity. It is the second condition on modal algebras that requires a generalization appropriate for intensional contexts. Just like an intensional partial order must broadly necessarily be, say, antisymmetric, an algebraic necessity must not just *in fact* commute with conjunction, i.e.,

$$\forall pq(M(p \wedge q) = Mp \wedge Mq).$$

Rather, this should hold as a matter of broad necessity. The higher-order identity in the second conjunct ensures that this will be so, without presupposing the notion of broad necessity.

Many key results from Bacon's theory of necessities have analogs for algebraic necessities, often with significantly streamlined proofs. Because algebraic necessities are defined purely equationally, we have $\mathtt{C} \vdash \mathsf{AlgNec}M \leftrightarrow (\mathsf{AlgNec}M = \top)$. Using this fact, we can straightforwardly show that $\Box$ is the broadest algebraic necessity, i.e.,

$$\mathtt{C} \vdash \mathsf{AlgNec}\Box \qquad\qquad \mathtt{C} \vdash \mathsf{AlgNec} \leq (\lambda M.\Box \leq M).$$

Indeed, $\top = \top \leftrightarrow \top$ is a theorem of $\mathtt{H}$, so by (Equiv) we get $\mathtt{C} \vdash \Box\top = \top$. Distribution over conjunction follows similarly from the classical tautologies $(p \wedge q) \rightarrow p$, $(p \wedge q) \rightarrow q$, $p \rightarrow q \rightarrow (p \wedge q)$ and $\beta\eta$-equivalence. To establish the second claim, note that $\mathtt{C} \vdash \mathsf{AlgNec}M \rightarrow p = \top \rightarrow Mp$. By (CEquiv) we thus have $\mathtt{C} \vdash \mathsf{AlgNec}M \rightarrow (\Box \leq M)$, which yields $\mathtt{C} \vdash \mathsf{AlgNec} \leq (\lambda M.\Box \leq M)$ by (Equiv).

This is no coincidence. Every result from Bacon's theory of necessities transfers to algebraic necessities, because algebraic necessities are the same thing as necessities in Bacon's sense.

Theorem 4.2. *In Classicism, to be an algebraic necessity just is to be a necessity in Bacon's sense:*

$$\mathtt{C} \vdash \mathsf{AlgNec} = \mathsf{Nec}.$$

Proof. ($\Rightarrow$) Assume $\mathsf{AlgNec}M$. First, we show that $\Box$ is a normal theory for M. The conditions $\mathsf{Taut}\Box$ and $\mathsf{MP}\Box$ hold because $\Box$ is a weak necessity, and $\mathsf{NecRule}\,\Box M$ holds because $p = \top$ implies $Mp = M\top = \top$. Note that $\mathsf{K}\Box M$ is just $\Box\mathsf{MP}M$, and $\mathsf{AlgNec}M$ entails $\mathsf{MP}M$. For if $Mp \wedge M(p \to q)$, then $M(p \wedge (p \to q))$ by distribution over conjunction. By $(p \wedge (p \to q)) = (p \wedge q)$ and distribution over conjunction again we get Mq. Using (CEquiv) twice gives $\Box\mathsf{MP}M$, noting the persistence of $\mathsf{AlgNec}M$. So $\Box$ is indeed a normal theory for M. Now, if p belongs to every normal theory of M, it must in particular satisfy $\Box$, and therefore must be true because $\Box$ is factive. This shows $\mathsf{WNec}M$, and $\mathsf{Nec}M$ by (CEquiv). Use (Equiv) to conclude.

($\Leftarrow$) Assume $\mathsf{Nec}M$. Since $\Box$ is a weak necessity and factive it follows that $\Box\mathsf{WNec}M$. This immediately gives $M\top = \top$ and $\mathsf{MP}M$. Assume $M(p \wedge q)$. Classicism proves both $(p \wedge q \to p)$ and $(p \wedge q \to q)$ so by (Equiv) and the fact that $\Box$ is the broadest necessity we have $M(p \wedge q \to p)$ and $M(p \wedge q \to q)$. By $\mathsf{MP}M$ both Mp and Mq follow. Conversely assume Mp and Mq. Use (Equiv) and the fact that $\Box$ is the broadest necessity to infer $M(p \to q \to (p \wedge q))$. Use $\mathsf{MP}M$ twice to conclude $M(p \wedge q)$. Conclude $\mathsf{AlgNec}M$ using (CEquiv) and $\mathsf{Nec} \leq \mathsf{AlgNec}$ using (Equiv). □

This result confirms that we are onto something. We have tried to formulate an object language notion of necessity from two different angles and ended up in the same place. There is a real logical joint here that deserves to be called the worldly analog of normal modal operators. Modal logics as deductive systems and modal algebras are two sides of the same coin, in the strong sense that every normal modal logic is sound and complete with respect to a class of modal algebras (Blackburn et al., 2001, Sec. 5.2). Theorem 4.2 shows that this connection between proof theory and algebraic semantics internalizes. In view of Theorem 4.2, I will stop using the terms 'algebraic necessity' and 'necessity in Bacon's sense' and speak simply of necessities.

4.2. **Intensional Filters.** To further expand the algebraic perspective on necessities, I give another characterization of necessities of a similarly algebraic flavor. A *filter* on a Boolean algebra is a non-empty collection of elements that is closed under meets and upward closed with respect to the underlying order. Modal operators on Boolean algebras correspond closely to filters. In a modal algebra with modal operator ■, the collection of elements a such that $\blacksquare a = 1$ is a filter, where 1 is the top element of the algebra. Conversely, any filter F determines a modal operator $\blacksquare_F$, putting $\blacksquare_F a = 1$ if a belongs to F, and 0 otherwise. Only modal operators that take just the values 0 and 1 are determined by filters in this way. In the internalized setting, by contrast, every necessity literally is what I shall call an *intensional filter*.[5]

Let us make this all precise.

Definition 4.3 (Intensional Filter)**.** An *intensional filter* is a property of propositions M that is necessarily non-empty, such that the relation two propositions bear

[5]Filters on a Boolean algebra also correspond one-to-one to congruences (Givant and Halmos, 2009, Chs. 17–18), and this correspondence can be internalized too: a necessity M determines the intensional analog of a congruence $\lambda pq.\, M(p \leftrightarrow q)$, and conversely such a relation R determines the necessity $\lambda p.\, Rp\top$. I do not pursue this here.

when they are both M entails the relation they bear when their conjunction is M, and M itself equals the property of being entailed by some M:

$$\mathsf{NE} := \lambda X.(\exists p\, Xp) = \top; \qquad \mathsf{CC} := \lambda X.(\lambda pq.Xp \wedge Xq) \leq (\lambda pq.X(p \wedge q));$$
$$\mathsf{CE} := \lambda X.X = (\lambda p.\exists q(Xq \wedge q \leq p)); \quad \mathsf{Filter!} := \lambda X.\mathsf{NE}X \wedge \mathsf{CC}X \wedge \mathsf{CE}X.$$

In the terminology of Section 3, intensional filters are the intensional counterparts of filters. The extension of an intensional filter at each world is a filter in the usual sense, but which proposition belong to the filter may vary across worlds.

The first, second, and third conjuncts internalize and intensionalize, respectively, the conditions that a filter be non-empty, closed under meets, and upward closed. Note that the second conjunct corresponds to the right-to-left direction of the second conjunct in the definition of an algebraic necessity. Each conjunct is equivalent to the claim that the corresponding extensional condition on filters, formalized in the obvious way, holds as a matter of broad necessity. The extension of an intensional filter necessarily satisfies the algebraic structure of a filter in the extensional sense. However, which propositions belong to the extension of an intensional filter is not fixed across modal space.

Our first characterization result relates necessities to intensional filters.

Proposition 4.4. *In Classicism, to be a necessity just is to be an intensional filter:*

$$\mathsf{C} \vdash \mathsf{Nec} = \mathsf{Filter!}.$$

Proof. ($\Rightarrow$) Assume $\mathsf{Nec}M$. The first two conjuncts of $\mathsf{Filter!}M$ obviously hold. For the third, obviously $Mp \rightarrow \exists q(Mq \wedge q \leq p)$. Conversely, if $\exists q(Mq \wedge q \leq p)$, then $p \wedge q = q$, so $M(p \wedge q)$. By distribution over conjunction, Mp. Conclude using (CEquiv).

($\Leftarrow$) Assume $\mathsf{Filter!}M$. By $(\exists p\, Mp) = \top$, the logical truth $\forall p(p \leq \top)$ and closure under entailment we have $M\top = \top$. We only need the converse of the second conjunct of $\mathsf{Filter!}M$, which follows because $(p \wedge q) \rightarrow q$ and $(p \wedge q) \rightarrow p$ are tautologies and intensional filters are closed under entailment. Conclude via (CEquiv). □

This simple result will prove to be a very useful tool in Section 5, where I study the algebraic structure of the family of necessities itself. Most importantly, I use it to establish a filter-theoretic characterization of the join of two necessities, which plays a crucial role in my proof that the intensional lattice of necessities is distributive.

5. The Intensional Lattice of Necessities

I now apply my algebraic characterization of necessities to study the (intensional) algebraic structure of the family of necessities itself. I review results due to Bacon and Zeng (2022) to the effect that necessities, like propositions, form an intensional bounded lattice with respect to entailment. I then apply my characterization of necessities as intensional filters to give a new characterization of arbitrary joins of persistent properties of necessities. This characterization yields a straightforward proof that the intensional lattice of necessities is *intensionally distributive*, in a sense to be specified. What is more, necessities satisfy an *infinite* intensional distributivity law that gives them the structure of what I call an *intensional Heyting algebra*.

5.1. **Bacon and Zeng's Results.** Recall from Section 3.1 that binary extensional and intensional meets and joins coincide, so I speak simply of the meet and join of two necessities.

Bacon and Zeng (2022) proved that, in Classicism, the necessities form an intensional bounded lattice with respect to entailment.[6] Since the conjunction of two necessities is always a necessity, the meet of two necessities always equals their conjunction:

$$\mathsf{C} \vdash \Box\forall MN(\mathsf{Nec}[M,N] \to \mathsf{Meet}(M \wedge_{t\to t} N)[M,N]\mathsf{Nec}).$$

We thus let

$$\sqcap := \lambda XY.X \wedge_{t\to t} Y.$$

However, the join of two necessities is in general weaker than their disjunction, for the disjunction of two necessities is not always a necessity. For example, it might happen that p is X-necessary and $p \to q$ is Y-necessary. Since necessities are closed under modus ponens, the join of X and Y should make q necessary, yet q might lack both X and Y. Bacon and Zeng (2022) observed that the join of two necessities X and Y can be characterized as the property of having every necessity entailed by the disjunction of X and Y:

$$\sqcup := \lambda XYp.\forall Z(\mathsf{Nec}Z \wedge (X \vee_{t\to t} Y) \leq Z \to Zp).$$

We then have

$$\mathsf{C} \vdash \Box\forall MN(\mathsf{Nec}[M,N] \to \mathsf{Join}(M \sqcup N)[M,N]\mathsf{Nec}).$$

So, Nec is an intensional lattice. In addition, it is *bounded*: as a matter of broad necessity, there are a *broadest* and a *narrowest* necessity. As we already remarked, the broadest necessity is $\Box$. On the other hand, the narrowest necessity is

$$\mathsf{Deg} := (\lambda p.\top).$$

The narrowest necessity Deg makes every proposition necessary (and no proposition possible).

Finally, the intensional lattice of necessities satisfies fairly strong completeness principles. Bacon and Zeng established that every rigid property of necessities has an extensional meet, and every *persistent* property that necessarily only applies to necessities has an intensional meet:

$$\mathsf{C} \vdash \Box\forall X(\mathsf{Rigid}X \wedge X \leq \mathsf{Nec} \to \exists M(\mathsf{Meet}MX\mathsf{Nec}));$$
$$\mathsf{C} \vdash \Box\forall X(\mathsf{Persistent}X \wedge X \leq \mathsf{Nec} \to \exists M(\mathsf{Meet!}MX\mathsf{Nec})).$$

In either case, the meet of the relevant property of necessities X coincides with the property of *having every necessity in* X:

$$\bigsqcap := \lambda Xp.\forall M(XM \to Mp)$$

What about joins? The claim that every *persistent* property of necessities has an *intensional* join is an easy corollary of Bacon and Zeng's result about meets:

$$\mathsf{C} \vdash \Box\forall X(\mathsf{Persistent}X \wedge X \leq \mathsf{Nec} \to \exists M(\mathsf{Join!}MX\mathsf{Nec})).$$

[6] Bacon and Zeng's results do not mention the notion of an intensional lattice and are couched in a grain-neutral framework. But the results I attribute to them here are simple corollaries of their results as they stated them.

When it exists, M coincides with the intensional meet of the property of being an intensional upper bound of X in Nec, which is itself persistent.

$$\bigsqcup := \lambda X.\bigsqcap \lambda Y.\mathsf{UB!}Y\,X\,\mathsf{Nec}$$

Thus, every rigid property of necessities has an extensional join, since rigid properties are persistent and the intensional join of a rigid property is also its extensional join.

5.2. **Characterization of Intensional Joins.** I now apply Proposition 4.4 to characterize intensional joins of necessities. The idea is simple. In extensional contexts, joins of filters are easy to characterize: the join of two filters is just the filter generated by their union, or the set of all elements in the underlying algebra that lie above the meet of two elements, one from each filter. We obtain a characterization of the join of two necessities simply by internalizing this construction.

Proposition 5.1. *In Classicism, the join of two necessities X and Y is the property of being entailed by the conjunction of some X-necessary proposition with some Y-necessary proposition:*

$$\mathsf{C} \vdash \Box\forall XY(\mathsf{Nec}[X,Y] \to X \sqcup Y = (\lambda p.\exists qr(Xq \wedge Yr \wedge (q \wedge r) \leq p))).$$

Proof. Let X and Y be necessities and let Z be the property of being entailed by the conjunction of some X-necessity and some Y-necessity as defined on the right-hand side above. We first show that Z is a necessity. Since $\top$ is entailed by everything, Z applies to something. If $p \leq q$ and Zp, there must be r, s with Xr and Ys such that $r \wedge s \leq p$. But then also $r \wedge s \leq q$, whence Zq. Last, if Zp and Zq, there must be p', p'', q', q'' with Xp', Yp'', Xq', and Yq'' such that $p' \wedge p'' \leq p$ and $q' \wedge q'' \leq q$. Since both X and Y are closed under conjunction, we have $X(p' \wedge q')$ and $Y(p'' \wedge q'')$. Note also that $(p' \wedge p'') \wedge (q' \wedge q'') \leq p \wedge q$, whence $Z(p \wedge q)$. Use (CEquiv) to conclude that Z is an intensional filter, hence a necessity.

Now, since $p \wedge \top \leq p$ and $Y\top$ are broadly necessary, Xp entails Zp. Likewise, Yp entails Zp. This shows $X \sqcup Y \leq Z$ via (CEquiv). Conversely, assume Zp, so that there are q, r with Xq and Yr such that $q \wedge r \leq p$. Since both X and Y entail $X \sqcup Y$ and necessities are closed under conjunction, we have $X \sqcup Y(q \wedge r)$, whence $X \sqcup Yp$ by closure under entailment. This shows $Z \leq X \sqcup Y$ via (CEquiv). Conclude by generalizing and necessitating. □

This characterization can be generalized to arbitrary intensional joins. The rough idea is that an arbitrary intensional join $\bigsqcup X$ of a persistent property that broadly necessarily only applies to necessities is the property of being entailed by some finite conjunction $p_0 \wedge p_1 \wedge \ldots \wedge p_n$ such that each p_i is necessary according to some necessity falling under X. To make that precise, we need to regiment the idea of being a conjunction of finitely many propositions.

To do that, let us take a detour through the notion of a *numerical* quantifier. A numerical quantifier, intuitively, is a quantifier of the form 'there are exactly n Xs', for some n. In higher-order logic, they can be recursively defined as follows:

$$\exists_0 := \lambda X.\neg\exists y Xy \qquad \mathsf{Suc} := \lambda QX.\exists x(Xx \wedge Q(\lambda y.Xy \wedge y \neq x)).$$

Using a standard Fregean trick, we then define the property of *being a numerical quantifier* as the property of having every property possessed by $\exists_0$ and closed

under Suc:

$$\mathsf{NumQ} := \lambda Q.\forall X(X\exists_0 \wedge \forall Q'(XQ' \rightarrow X(\mathsf{Suc}Q')) \rightarrow XQ).$$

We then let a *finite property* be one to which some numerical quantifier applies:

$$\mathsf{Fin} := \lambda X.\exists Q(\mathsf{NumQ}Q \wedge QX).$$

A finite conjunction is nothing but the extensional meet of a finite property of propositions. We know that, in Classicism, the (intensional and extensional) meet of a *rigid* property of propositions X is just $\forall p(Xp \rightarrow p)$ (and its join is $\exists p(Xp \wedge p)$). So, if we could show that any finite property of propositions is coextensive with a rigid property, then we would have our notion of a finite conjunction.

Predictably, this can be done. Say that a property X *rigidifies* a property Y when X is rigid and coextensive with Y:

$$\mathsf{Rigidifies} := \lambda XY.\mathsf{Rigid}X \wedge \forall z(Xz \leftrightarrow Yz).$$

We can show that every finite property is rigidified by something by showing that the property of *only applying to properties that have a rigidifier* is possessed by $\exists_0$ and closed under Suc, and so must be shared by all numerical quantifiers.

Proposition 5.2. *In Classicism, every finite property of propositions has a rigidifier:*

$$\mathsf{C} \vdash \Box\forall X(\mathsf{Fin}X^{t\rightarrow t} \rightarrow \exists Y(\mathsf{Rigidifies}YX)).$$

Proof. We show that $F := \lambda Q.\forall X^{t\rightarrow t}(QX \rightarrow \exists Y(\mathsf{Rigidifies}YX))$ holds of $\exists_0$ and is closed under Suc, so that it holds of every numerical quantifier. The base case $F\exists_0$ holds because $\lambda p.\bot$ is rigid and coextensive with any property with empty extension. If FQ and $\mathsf{Suc}QX$, pick p with Xp and $Q(\lambda q.Xq \wedge q \neq p)$, and let Y rigidify $\lambda q.Xq \wedge q \neq p$. Then $\lambda r.Yr \vee r = p$ is rigid and coextensive with X. □

In view of Proposition 5.2, quantification over finite properties of propositions can, without loss of generality, be replaced with quantification over *rigid* finite properties of propositions. So, by the *conjunction*, or meet, of a finite property of propositions X we will mean the proposition $\forall p(Yp \rightarrow p)$, where Y is a rigidifier of X. Likewise, by the *disjunction*, or join, of a finite property of propositions X we will mean the proposition $\exists p(Yp \wedge p)$, where Y is a rigidifier of X. These propositions always give the extensional and intensional meets and joins of any rigid property of propositions.

Using a similar inductive proof, we can show that all necessities are closed under finite conjunctions.

Proposition 5.3. *In Classicism, every necessity applies to the greatest lower bound of any finite, rigid property of propositions that only applies to propositions that are necessary in that sense of necessity:*

$$\mathsf{C} \vdash \Box\forall M(\mathsf{Nec}M \rightarrow \forall X(\mathsf{Fin}X \wedge \mathsf{Rigid}X \wedge X \subseteq M \rightarrow M(\forall p(Xp \rightarrow p)))).$$

Proof. By the same kind of induction. If $\exists_0 X$, then $\forall p(Xp \rightarrow p) = \top$. If $\mathsf{Suc}QX$, pick p with Xp and $Q(\lambda q.Xq \wedge q \neq p)$, and let Y rigidify $\lambda q.Xq \wedge q \neq p$. Then Mp, and $M(\forall q(Yq \rightarrow q))$ by the induction hypothesis, since numerical quantifiers cannot distinguish between coextensive properties. As X is rigid, it is coextensive, hence identical, with $\lambda r.r = p \vee Yr$, so $\forall q(Xq \rightarrow q) = p \wedge \forall q(Yq \rightarrow q)$, which is M-necessary by closure under conjunction. □

We need one more definition. When $X : (t \to t) \to t$ and $Y : t \to t$, we say that X *covers* Y when every Y proposition possesses one of the X-properties:

$$\mathsf{Covers} := \lambda X^{(t\to t)\to t} Y^{t\to t}.\forall p(Yp \to \exists Z(XZ \wedge Zp)).$$

When, in particular, X is a property of necessities, to say that X covers Y means that every Y-proposition is necessary according to some necessity possessing X.

We are now ready to generalize Proposition 5.1 to arbitrary joins.

Proposition 5.4. *In Classicism, the join of a persistent property X that broadly necessarily only applies to necessities is the property of being entailed by the conjunction of some rigid finite property of propositions that X covers:*

$$\mathsf{C} \vdash \Box\forall X(\mathsf{Persistent}X \wedge X \leq \mathsf{Nec} \to \bigsqcup X = (\lambda p.\exists Z(\mathsf{Fin}Z \wedge \mathsf{Rigid}Z \wedge \mathsf{Covers}XZ \wedge \forall q(Zq \to q) \leq p))).$$

Proof. Fix X and let W be the property on the right-hand side of the identity above. First, we show W is a necessity. If X applies to nothing, then $\lambda p.\bot$ is finite, rigid, vacuously covered by X, and such that $\forall p(\bot \to p) = \top$, which shows $W\top$. Otherwise, $\Box$ is finite, rigid, covered by X and such that $\forall p(\Box p \to p) = \top$, which again shows $W\top$. For closure under conjunction, clearly if $r \leq p \wedge q$ then $r \leq p$ and $r \leq q$. Assume Wp and Wq, witnessed by Y, Z finite, rigid, covered by X and such that $\forall r(Yr \to r) \leq p$ and $\forall r(Zr \to r) \leq q$. Then $Y \vee_{t\to t} Z$ is also finite, rigid and covered by X, and clearly $\forall r((Yr \vee Zr) \to r) \leq p \wedge q$. We have shown that $W(p \wedge q) \leftrightarrow Wp \wedge Wq$; (CEquiv) gives the desired identity. So, W is a necessity.

Now assume Mp for some M with XM. Then $\lambda q.p = q \vee q = \top$ is a finite, rigid property covered by X whose greatest lower bound, namely p, entails p. That shows Wp. Using (CEquiv) this shows that W is an intensional upper bound of X, whence $\bigsqcup X \leq W$.

Conversely, assume Wp. Then there is a finite, rigid property Z covered by X such that $\forall q(Zq \to q) \leq p$. This implies $Z \subseteq \bigsqcup X$: for Zr implies that there is some necessity M with XM such that Mr, and $M \leq \bigsqcup X$. But necessities are closed under finite conjunction in the sense of Proposition 5.3, so $\bigsqcup X(\forall r(Zr \to r))$ holds. Since $\forall q(Zq \to q) \leq p$ we must also have $\bigsqcup Xp$. By (CEquiv) we may conclude $W \leq \bigsqcup X$. □

5.3. **Distributivity and Heyting Algebra Structure.** A lattice is called *distributive* when meets and joins distribute over one another, the same way conjunction and disjunction do. In our intensional setting, the right notion to work with is that of an *intensionally distributive property* (with respect to entailment). It can be straightforwardly formulated as follows:

$$\mathsf{Distr!} := \lambda X.\Box\forall MNO(X[M,N,O] \to M \sqcap (N \sqcup O) = (M \sqcap N) \sqcup (M \sqcap O)).$$

There is a natural infinitary strengthening of this property, which we can call *infinite* intensional distributivity:

$$\mathsf{Distr!}^\infty := \lambda Y.\Box\forall MX(YM \wedge X \leq Y \wedge \mathsf{Persistent}X \to M \sqcap \bigsqcup X = \bigsqcup\nolimits_{XZ} M \sqcap Z),$$

where

$$\bigsqcup\nolimits_{XZ} M \sqcap Z := \bigsqcup \lambda W.\exists Z(XZ \wedge W = M \sqcap Z).$$

While not explicitly appealing to my notions of intensional distributivity, Bacon and Zeng (2022, p. 177) raised the question of whether necessities form an intensionally distributive lattice, but left it open. In this section, I apply Proposition 5.4 to answer this question in the affirmative. In fact, I establish the stronger result that the intensional lattice of necessities is infinitely intensionally distributive.[7] It follows that necessities have the structure of an *intensional Heyting algebra*, in a sense to be specified.

While intensional distributivity is clearly a consequence of infinite intensional distributivity, it is instructive to go through the simpler proof of the former first.

Theorem 5.5. *In Classicism, to be a necessity is an intensionally distributive property:*

$$\mathsf{C} \vdash \mathsf{Distr!Nec}.$$

Proof. Let M, N, O be as required by the antecedent of Distr!Nec. The right-to-left direction holds in any lattice. For the converse, assume $(M \sqcap (N \sqcup O))\, p$. Then Mp and $(N \sqcup O)p$. So, there must be q, r with Nq and Or such that $q \wedge r \leq p$. Since necessities are closed under entailment we have $M(p \vee q)$, $N(p \vee q)$, $M(p \vee r)$ and $O(p \vee r)$, so $(M \sqcap N)(p \vee q)$ and $(M \sqcap O)(p \vee r)$. But Classicism proves the identity $(p \vee q) \wedge (p \vee r) = p \vee (q \wedge r)$, which entails p because both disjuncts do. Consequently, $((M \sqcap N) \sqcup (M \sqcap O))p$, as desired. Use (CEquiv), then necessitate and generalize to conclude. □

In the proof just given, we derive the distributivity of meets over binary joins of necessities by appealing to the distributivity of disjunction over binary conjunctions of propositions. The general case goes through because there is nothing special about the binary case: disjunction distributes over any finite conjunction, not just binary ones. We just had to be a little careful in properly formalizing finite conjunctions.

Theorem 5.6. *In Classicism, to be a necessity is an infinitely intensionally distributive property:*

$$\mathsf{C} \vdash \mathsf{Distr!^{\infty}Nec}.$$

Proof. The ($\Leftarrow$) direction holds in any lattice. For ($\Rightarrow$) let X and M be as required by the antecedent of Distr!$^{\infty}$Nec. Assume $(M \sqcap \bigsqcup X)\, p$. Then Mp and $\bigsqcup Xp$. So, there must be some finite, rigid Y covered by X such that $\forall q(Yq \to q) \leq p$. Now, consider the property of being a disjunction of p with some Y-proposition,

$$O' := \lambda q.\exists r(Yr \wedge (q = p \vee r)).$$

It is clearly finite. Let O be a rigidifier of O'. Notice that whenever Or holds, we have both Mr (because Mp and $p \leq r$) and Nr for some N with XN (because $r = p \vee r'$ for some r' with Yr', and X covers Y). So, Oq implies $(M \sqcap N)q$ for some N with XN. This means that O is covered by $\lambda N.\exists N'(XN' \wedge N = M \sqcap N')$, which $\bigsqcup_{XN} M \sqcap N$ is the join of. So, we are done if we can show that $\forall q(Oq \to q) \leq p$. This is indeed the case. Note that $\forall q(Oq \to q) = p \vee \forall q(Yq \to q)$. For the right-to-left entailment, observe that Classicism proves $p \vee \forall q(Yq \to q) = \forall q(Yq \to p \vee q)$,

[7] It is well known that the lattice of filters on a Boolean algebra is a frame, i.e., satisfies the infinite distributive law, see, e.g., Johnstone (1982, Chap. II, Sec 2.11). Proving that Nec is infinitely intensionally distributive amounts to generalizing this fact to an intensional setting.

and the right-hand side clearly entails $\forall q(Oq \to q)$. Conversely, assume $\forall q(Oq \to q)$ and suppose $\neg p$. Take any q with Yq. We know $O(p \vee q)$, so $p \vee q$. That can only be because q is true. Necessitating this reasoning shows the left-to-right entailment. But now $p \vee \forall q(Yq \to q) \leq p$ clearly holds, since both p and $\forall q(Yq \to q)$ entail p. Conclude as in the finite case. □

A *Heyting algebra* is a bounded distributive lattice in which, for any two elements a and b, there is a *pseudocomplement* of a relative to b, a weakest element $a \to b$ such that the meet of a and $a \to b$ is below b (Esakia, 2019, Sec. 2.1). Heyting algebras are the algebraic semantics of intuitionistic propositional logic, in the same way that Boolean algebras are the algebraic semantics of classical propositional logic. Relative pseudocomplements are used to interpret conditionals: the very definition of relative pseudocomplements can be seen as an algebraization of the introduction and elimination rules for the conditional.

Like we did for other algebraic properties, it is natural to introduce the concept of an *intensional Heyting algebra*. I will not define this concept in full generality; I will only define what it would take for necessities to form an intensional Heyting algebra. An *intensional relative pseudocomplement* operation on necessities would be a binary operation on necessities satisfying:

$$\begin{aligned}\mathsf{RPseudo!} := \lambda R.\Box\forall MN(\mathsf{Nec}[M,N] \to \mathsf{Nec}(RMN)) \wedge \\ \Box\forall MNO(\mathsf{Nec}[M,N,O] \to (M \sqcap O \leq N \leftrightarrow O \leq RMN)).\end{aligned}$$

Necessities form an intensional Heyting algebra if an intensional relative pseudocomplement operation on necessities exists.

In extensional contexts, every complete lattice satisfying the infinite distributive law is a Heyting algebra (Johnstone, 1982, II.1.1). This generalizes to our intensional notions, allowing us to prove:

Theorem 5.7. *In Classicism, necessities form an intensional Heyting algebra, where the intensional relative pseudocomplement operation is given by*

$$\sqsupset := \lambda MN.\bigsqcup(\lambda O.\mathsf{Nec}O \wedge M \sqcap O \leq N).$$

That is, $\mathsf{C} \vdash \mathsf{RPseudo!}(\sqsupset)$.

Proof. Fix necessities M, N. Let $F := \lambda O.\mathsf{Nec}O \wedge M \sqcap O \leq N$. The property F is persistent and entails Nec. So, by Proposition 5.4, $M \sqsupset N$ is its intensional join in Nec.

Now let O be a necessity. If $M \sqcap O \leq N$, then FO, so $O \leq M \sqsupset N$. Conversely, if $O \leq M \sqsupset N$, then $M \sqcap O \leq M \sqcap (M \sqsupset N)$. By Theorem 5.6, $M \sqcap (M \sqsupset N) = \bigsqcup_{FO'}(M \sqcap O')$. But N is an intensional upper bound of the property of being equal to $M \sqcap O'$ for some O' with FO', so $M \sqcap (M \sqsupset N) \leq N$. Hence $M \sqcap O \leq N$. Generalize and necessitate to conclude. □

The fact that $\lambda O.\mathsf{Nec}O \wedge M \sqcap O \leq N$ is a persistent property of necessities is a crucial ingredient in the proof, else we would not be able to show that its intensional join exists.

6. Pseudocomplements and Defective Necessities

Once we have relative pseudocomplements, we can define the *absolute* pseudocomplement of a necessity M as its pseudocomplement relative to $\Box$, the bottom element of our intensional lattice:

$$\overline{M} := M \sqsupset \Box.$$

We can think of $\overline{M}$ as the (intuitionistic) *negation* of a necessity M, in the same way that we can define the negation of a proposition p as $p \to \bot$.

A necessity M is said to satisfy the *Leibniz Biconditionals* when to be M-necessary is to be true at all M-possible worlds. The notions of an M-possible world and that of truth at a world can be precisely defined in Classicism (see Bacon, 2018; Fritz, 2023), but we do not need to get into the details here. The pseudocomplement of a necessity M that does satisfy the Leibniz Biconditionals is the property of being true at all *M-impossible worlds*. This modality has been studied model-theoretically in, e.g., Humberstone (1983) and Gargov et al. (1987). But Classicism does not guarantee that all necessities satisfy the Leibniz Biconditionals (see Bacon and Dorr, 2024, Sec. 2.2). The claim that they do carries substantive metaphysical commitments. The pseudocomplement operation generalizes the same idea to all necessities, regardless of whether they satisfy the Leibniz Biconditionals. From truth at all M-impossible worlds, we generalize to *broadly necessarily being entailed by all M-impossible propositions*.

Proposition 6.1. *In Classicism, the pseudocomplement of a necessity M is the property of broadly necessarily being entailed by every M-impossible proposition—that is, of being an upper bound of the property of being M-impossible:*

$$\mathsf{C} \vdash \Box \forall M(\mathsf{Nec}M \to \overline{M} = (\lambda p.\mathsf{UB}p(\lambda q.M\neg q)))$$

Proof. Let M be a necessity. Note that $\top$ is entailed by everything, and a conjunction is an upper bound of a property iff each conjunct is. Using (CEquiv), this establishes that $\lambda p.\mathsf{UB}p(\lambda q.M\neg q)$ is a necessity.

Next, I claim $M \sqcap \lambda p.\mathsf{UB}p\lambda q.M\neg q \leq \Box$. For assume Mp and $\mathsf{UB}p\lambda q.M\neg q$. Then p must be entailed by its own negation, which is only true of $\top$. (CEquiv) establishes the entailment. By the properties of $\sqsupset$ this implies $\lambda p.\mathsf{UB}p\lambda q.M\neg q \leq \overline{M}$. For the converse, note first that $M \sqcap \overline{M} = \Box$ by the properties of $\sqsupset$. Assume $\overline{M}p$ and $M\neg q$. Then both $\overline{M}(p \vee \neg q)$ and $M(p \vee \neg q)$ hold, so we must have $p \vee \neg q = \top$. This is to say $q \leq p$. Generalizing, $\mathsf{UB}p\lambda q.M\neg q$. Conclude with (CEquiv). $\Box$

Pseudocomplements in Nec do not in general behave classically. But some do. Call a necessity M *Boolean* when it satisfies the law of excluded middle, $M \sqcup \overline{M} = \mathsf{Deg}$, and *regular* when it is identical to its double pseudocomplement, $\overline{\overline{M}} = M$:

$$\mathsf{Boolean} := \lambda M.\, M \sqcup \overline{M} = \mathsf{Deg}; \qquad \mathsf{Regular} := \lambda M.\, \overline{\overline{M}} = M.$$

Every Boolean necessity is regular, but what we have established about Nec so far leaves open whether there are regular, non-Boolean necessities. As it turns out, Classicism is consistent with the existence of necessities that are neither regular nor Boolean, as well as with the existence of regular, but non-Boolean necessities.

In the remainder of the article, I describe the conditions under which necessities are Boolean or regular. I will show that being Boolean and being regular correspond to two different senses in which a necessity can be said to be *defective*. For

illustration, take the example of the necessity *having probability 1* from the introduction. As I observed earlier, there are at least two senses in which this necessity is defective.

First, it is not *principal*: for every proposition that has probability 1, we can find a strictly stronger proposition that also has probability 1; there is no maximally strong proposition with probability 1. Suppose I throw a dart at a board, and consider, for each point x of the board, the proposition that the dart does not hit x. Each of these propositions has probability 1, while the proposition that the dart hits no point of the board has probability 0. Now, let p be any proposition with probability 1. Then there must be a point x such that p does not entail that the dart does not hit x, since otherwise p would entail that the dart hits no point of the board, which has probability 0. Given one such x, the conjunction of p with the proposition that the dart does not hit x is strictly stronger than p and has probability 1.

The second defect in the necessity *having probability 1* is that it is not *infinitely closed*: while finitary consequences of propositions with probability 1 also have probability 1, there are infinitary consequences of propositions with probability 1 that fail to have probability 1. In the same scenario as before, the conjunction of the propositions that the dart does not hit x, for each point x, is an infinitary consequence of those propositions, each of which has probability 1. But clearly, it has probability 0, not 1.

I regiment the notions of principal and infinitely closed necessities and characterize them in terms of the classical principles on pseudocomplements that they correspond to: being Boolean corresponds to being principal (excluded middle), while being regular corresponds to being infinitely closed (double negation elimination).

6.1. **Boolean Necessities.** Call a necessity M *principal* when it is *generated* by a single proposition, in the sense that, broadly necessarily, there is a proposition p that entails precisely the M-necessary propositions:

$$\mathsf{Princ} := \lambda M.\Box\exists p(\forall q(Mq \leftrightarrow p \leq q)).$$

Equivalently, a necessity is principal when, broadly necessarily, it has an extensional meet and makes it necessary:

$$\mathsf{C} \vdash (\mathsf{Nec} \wedge \mathsf{Princ}) = \lambda M.\mathsf{Nec}M \wedge \Box\exists p(\mathsf{Meet}pM \wedge Mp).$$

Indeed, if M is principal, then its extensional meet is whichever proposition generates it. And conversely, any necessity that broadly necessarily makes its extensional meet necessary must be generated by the same.

Intuitively, then, a principal necessity is one that admits a single strongest necessary proposition, as a matter of broad necessity. This proposition can, but does not have to, be the same across worlds, so to speak. For example, *being entailed by Alice's favorite proposition* is a principal necessity, even if Alice does not have the same favorite proposition across all worlds. Any property of the form

$$\Box_p := \lambda q.p \leq q,$$

for any proposition p, is a principal necessity. In particular, $\Box = \Box_\top$ and $\mathsf{Deg} = \Box_\bot$, so both are principal.

An easy way to construct concrete examples of non-principal necessities is to take a collection of propositions closed under conjunction and entailment that has

a greatest lower bound, which however cannot be expressed as the *finite* conjunction of propositions in the collection. Here is one such example. I point the reader to Bacon (2024) for details about the semantics of higher-order logic.

Example 6.2. Construct a model of higher-order logic as follows. We think of natural numbers as worlds. Let the domain of propositions $\mathbf{A}^t$ consist of all sets of natural numbers, let the domain of individuals be arbitrary and let higher-order domains $\mathbf{A}^{\sigma\to\tau}$ consist of all functions from $\mathbf{A}^\sigma$ to $\mathbf{A}^\tau$. We interpret Boolean connectives as set-theoretic operations and quantifiers as infinite intersections and unions. Consider now the function $\mathbf{f}$ that maps a set of natural numbers to $\mathbb{N}$ iff it is cofinite and includes all even numbers, and to $\varnothing$ otherwise. In our model, this function is a necessity: the intersection of any two such sets is again cofinite and includes all even numbers, as is any superset thereof. But the intersection of all cofinite sets containing all even numbers is the set of even numbers itself, which is not cofinite. So, $\mathbf{f}$ is not principal: while its extensional meet exists, $\mathbf{f}$ does not make it necessary.

As our first application of Proposition 6.1, I show that the Boolean necessities are precisely the principal ones. We first apply Proposition 6.1 to characterize the pseudocomplement of a principal necessity, then use it to derive the desired identity.

Proposition 6.3. *In Classicism, the pseudocomplement of a principal necessity M is the property of being entailed by the negation of its greatest lower bound:*

$$\mathsf{C} \vdash \Box\forall M(\mathsf{Nec}M \wedge \mathsf{Princ}M \to \overline{M} = (\lambda q.\exists p(\mathsf{Meet}pM \wedge \neg p \leq q))).$$

Proof. Let M be a principal necessity and p its meet. We know from Proposition 6.1 that $\overline{M}$ equals the property of being entailed by every M-impossible proposition. But if p is the meet of M—that is, the strongest M-necessary proposition—then $\neg p$ must be the weakest M-impossible proposition. So, to be entailed by $\neg p$ is to be entailed by every M-impossible proposition. □

Theorem 6.4. *In Classicism, to be a Boolean necessity just is to be a principal necessity:*

$$\mathsf{C} \vdash (\mathsf{Nec} \wedge \mathsf{Boolean}) = (\mathsf{Nec} \wedge \mathsf{Princ}).$$

Proof. Assume M is a principal necessity. Let p be the greatest lower bound of M. By Proposition 6.3, $\overline{M}\neg p$, so because $p \wedge \neg p = \bot$, we have $(M \sqcup \overline{M})\bot$ by Proposition 5.1. Necessitating this reasoning shows $(M \sqcup \overline{M}) = \mathsf{Deg}$.

Conversely, let M be a Boolean necessity. Then $M \sqcup \overline{M} = \mathsf{Deg}$. By Proposition 5.1 and the fact that $\mathsf{Deg}\bot$, it follows that there must be some p, q such that Mp and $\overline{M}q$ and $p \wedge q = \bot$. This implies $q \leq \neg p$, so that $\overline{M}(\neg p)$. I claim that M is generated by p. Indeed, take any r with Mr. Then $M(\neg p \vee r)$ and $\overline{M}(\neg p \vee r)$ by closure under entailment. Since $M \sqcap \overline{M} = \Box$, that means $\neg p \vee r = \top$, which is to say $p \leq r$. Necessitating this reasoning yields the left-to-right entailment. □

Proposition 6.3 draws a nice connection between pseudocomplementation and negation: the pseudocomplement of a Boolean necessity is the necessity generated by the negation of its greatest lower bound. This connection is not unique to pseudocomplementation; it extends to meets and joins as well, relating meets to disjunctions and joins to conjunctions. The *meet* of two Boolean necessities is just

the necessity generated by the *disjunction* of their greatest lower bounds, and the *join* of two Boolean necessities is just the necessity generated by the *conjunction* of their greatest lower bounds. This shows, transparently, that Boolean necessities form a Boolean subalgebra of the algebra of necessities.

Proposition 6.5. *In Classicism, if M, N are Boolean necessities, then $M \sqcap N$ is the property of being entailed by the disjunction of the greatest lower bounds of M and N, and $M \sqcup N$ is the property of being entailed by their conjunction. Consequently, both are Boolean.*

$$\begin{aligned}\mathsf{C} \vdash \Box\forall MN((\mathsf{Nec} \wedge \mathsf{Princ})[M,N] \rightarrow \\ (M \sqcap N = (\lambda p.\exists qr(\mathsf{Meet}qM \wedge \mathsf{Meet}rN \wedge (q \vee r) \leq p)))) \\ \mathsf{C} \vdash \Box\forall MN((\mathsf{Nec} \wedge \mathsf{Princ})[M,N] \rightarrow \\ (M \sqcup N = (\lambda p.\exists qr(\mathsf{Meet}qM \wedge \mathsf{Meet}rN \wedge (q \wedge r) \leq p))))\end{aligned}$$

Proof. Let M, N be Boolean necessities. For meets, observe that p is entailed by both the greatest lower bound of M and that of N iff it is entailed by their disjunction. For joins, if $(M \sqcup N)p$, by Proposition 5.1 there are q, r with Mq and Nr such that $q \wedge r \leq p$. The greatest lower bound of M entails q and the greatest lower bound of N entails r, so their conjunction entails p. The converse follows similarly from Proposition 5.1. □

6.2. **Regular Necessities.** Let us now turn to regular necessities, those equal to their double pseudocomplement. Obviously, for any necessity M, broadly necessarily, if Mp, then p is entailed by every lower bound of M. The converse need not be true. When it is, we call M a *cut*:[8]

$$\mathsf{Cut} := \lambda M.\Box\forall p(Mp \leftrightarrow \forall q(\mathsf{LB}qM \rightarrow q \leq p)).$$

Intuitively, a cut is a necessity that can be uniquely reconstructed from its lower bounds: anything entailed by all lower bounds is already necessary according to the cut.

Any principal necessity is a cut:

$$\mathsf{C} \vdash (\mathsf{Nec} \wedge \mathsf{Princ}) \leq (\mathsf{Nec} \wedge \mathsf{Cut}).$$

For if M necessarily has a greatest lower bound p, then q is entailed by p iff it is entailed by every lower bound of M. However, intuitively a necessity can be a cut without being principal if it possibly lacks an extensional meet. Consider, for example, the Boolean algebra of finite and cofinite subsets of $\mathbb{N}$. Let E be the function mapping every cofinite subset of $\mathbb{N}$ that contains every even number to $\mathbb{N}$ (the top element of the algebra), and everything else to $\varnothing$. Then E is a necessity. It is also a cut: the lower bounds of E are precisely the finite subsets of even numbers, and a set contains (is entailed by) every finite subset of even numbers iff it contains the set of even numbers itself. The set of even numbers itself does not live in our algebra—only its cofinite supersets do, which are all mapped by E to the top element $\mathbb{N}$. However, E is not principal. Because, as we just saw, its lower

[8]The name 'cut' is inspired by the literature on completions of posets, where a set is called a cut if it equals the set of all upper bounds of all its lower bounds. These sets are also called *Dedekind–MacNeille closed*. See, e.g., Davey and Priestley (2002, 7.38).

bounds are precisely the finite subsets of even numbers, E has no extensional meet, and so cannot be principal.

This argument is not a *proof* that, in the object language, not every cut is a principal necessity. To prove that, we would have to show that the model we relied on can be extended to a full model of Classicism that supports the same reasoning. This particular example does not seem to admit such an extension.[9] To actually prove that not every cut is principal, we can consider countermodels to the principle of *Boolean Completeness* at type $t \to t$, which says that every property of propositions has an extensional meet:

$$\forall X^{t\to t}\exists p \mathsf{Meet} p X. \tag{Boolean Comp}$$

A model of Classicism that refutes (Boolean Comp) must contain a property F that lacks an extensional meet. One can then use F to construct a non-principal cut, obtained by closing F under the *multi-premise entailment* relation defined below. Through such an argument we thus establish:

Theorem 6.6. *It is consistent with Classicism that not every cut is a principal necessity:*

$$\mathsf{C} \nvdash \forall X(\mathsf{Nec}X \to \mathsf{Cut}X \to \mathsf{Princ}X).$$

While it is known that (Boolean Comp) is independent of Classicism, countermodels to (Boolean Comp) are less than straightforward to construct. I offer one in Appendix A and explain how to construct a non-principal cut in detail.

To say that p is entailed by every lower bound of M is a natural regimentation of the idea that M, a potentially infinite property of propositions, *multi-premise entails* p. Indeed, Bacon (2024, Def. 7.7) defines multi-premise entailment exactly this way:

$$\trianglelefteq := \lambda X^{t\to t} p.\forall q(\mathsf{LB}qX \to q \leq p).$$

Every cut is infinitely closed, in the sense that it makes necessary any multi-premise consequence of propositions it makes necessary:

$$\mathsf{C} \vdash \Box\forall M(\mathsf{Nec}M \wedge \mathsf{Cut}M \to \Box\forall X p(X \subseteq M \wedge X \trianglelefteq p \to Mp)).$$

What is more, it turns out that the regular necessities are precisely the cuts. If we compute the double pseudocomplement of a necessity M using Proposition 6.1, we get exactly the property of being entailed by every lower bound of M, which implies the desired identity.

Theorem 6.7. *In Classicism, the double pseudocomplement of a necessity M is the property of being entailed by every lower bound of M:*

$$\mathsf{C} \vdash \Box\forall M(\mathsf{Nec}M \to \overline{\overline{M}} = (\lambda p.\forall q(\mathsf{LB}qM \to q \leq p))).$$

Consequently, to be a regular necessity just is to be a cut:

$$\mathsf{C} \vdash (\mathsf{Nec} \wedge \mathsf{Regular}) = (\mathsf{Nec} \wedge \mathsf{Cut}).$$

[9] When constructing a model of Classicism based on it that shows what we hope to show, we would have to put the function E in the domain at type $t \to t$. But if we do that, the set of even numbers would belong to the domain at type t, because it would equal the proposition that every E-proposition is true. Still, I suspect that some variant of the argument just sketched can be made to work.

Proof. Let M be a necessity. By Proposition 6.1, $\overline{\overline{M}}$ equals the property of being entailed by every proposition whose negation is entailed by every M-impossible proposition: $\lambda p.\mathsf{UB}p\lambda q.\mathsf{UB}\neg q\lambda r.M\neg r$. But a proposition whose negation is entailed by every M-impossible proposition is just a proposition that entails every M-necessary proposition, so $\overline{\overline{M}} = (\lambda p.\mathsf{UB}p\lambda q.\mathsf{LB}qM)$. Spelling out the right-hand side gives $\lambda p.M \trianglelefteq p$. □

Thus, the double pseudocomplementation operation coincides with the operation of closing a necessity under multi-premise entailment.

Regular necessities also form an intensional Boolean algebra,[10] though this need not be a subalgebra of Nec. The meet of two regular necessities is always regular, as double pseudocomplementation commutes with finite meets. The pseudocomplement of a regular necessity is also regular. However, the join of two regular necessities need not be regular. Intuitively, this is so because the lower bounds of $M \sqcup N$ are precisely those propositions that entail all finite conjunctions of M-necessities with N-necessities. But only the consequences of such finite conjunctions are necessary according to $M \sqcup N$.

To verify this, it is helpful to appeal to the fact that double pseudocomplementation is a *closure operator*: an increasing, monotonic and idempotent operator.

Proposition 6.8. *In Classicism, $\bar{\bar{\cdot}}$ is a closure operator on* Nec*:*

$$\mathsf{C} \vdash \Box\forall M(\mathsf{Nec}M \to M \leq \overline{\overline{M}})$$

$$\mathsf{C} \vdash \Box\forall MN(\mathsf{Nec}M \wedge \mathsf{Nec}N \wedge M \leq N \to \overline{\overline{M}} \leq \overline{\overline{N}})$$

$$\mathsf{C} \vdash \Box\forall M(\mathsf{Nec}M \to \overline{\overline{\overline{M}}} = \overline{\overline{M}}).$$

Proof. Double pseudocomplementation is a closure operator in any Heyting algebra (Johnstone, 1982, I.1.13). In our setting, all identities above are also easily established by reasoning about lower bounds. Every M-necessity is entailed by every lower bound of M. If $M \leq N$, then every lower bound of N is a lower bound of M. Last, $\overline{\overline{\overline{M}}}p$ holds iff p is entailed by all lower bounds of $\overline{\overline{M}}$, and these are precisely the lower bounds of M. □

Proposition 6.9. *In Classicism, the regular necessities form a Boolean algebra. Meets and complements are given by $\sqcap$ and $\bar{\cdot}$ respectively, while the join of M and N is given by $\overline{\overline{M \sqcup N}}$:*

$$\mathsf{C} \vdash \Box\forall MN((\mathsf{Nec} \wedge \mathsf{Regular})[M,N] \to \mathsf{Regular}(M \sqcap N) \wedge \mathsf{Regular}(\overline{M}))$$

$$\mathsf{C} \vdash \Box\forall MN((\mathsf{Nec} \wedge \mathsf{Regular})[M,N] \to \mathsf{Join}\overline{\overline{(M \sqcup N)}}[M,N](\mathsf{Nec} \wedge \mathsf{Regular}))$$

$$\mathsf{C} \vdash \Box\forall MNO((\mathsf{Nec} \wedge \mathsf{Regular})[M,N,O] \to M \sqcap \overline{\overline{(N \sqcup O)}} = \overline{\overline{(M \sqcap N) \sqcup (M \sqcap O)}})$$

$$\mathsf{C} \vdash \Box\forall M(\mathsf{Nec}M \wedge \mathsf{Regular}M \to \overline{\overline{M \sqcup \overline{M}}} = \mathsf{Deg}).$$

[10] This is an intensional analog of Glivenko's theorem that the regular elements of a Heyting algebra form a Boolean algebra (Glivenko, 1929).

Proof. Let M, N be regular necessities. For meets, simply observe that the lower bounds of $M \sqcap N$ are exactly the propositions that are lower bounds of all propositions that are both M- and N-necessary, so anything entailed by these is already necessary according to both M and N. For joins, obviously $\overline{\overline{M \sqcup N}}$ is regular, and because $\overline{\overline{\cdot}}$ is increasing it is also an upper bound of M and N. That it is the least such follows from the monotonicity and idempotence of $\overline{\overline{\cdot}}$.

Distributivity follows from Distr!Nec plus the fact that $\overline{\overline{\cdot}}$ commutes with finite meets.[11] That gives us $\overline{\overline{M}} \sqcap \overline{\overline{(N \sqcup O)}} = \overline{\overline{(M \sqcap N) \sqcup (M \sqcap O)}}$, and the regularity of M gives the desired identity.

For complements, assume p is entailed by every lower bound of $\overline{M}$. If q is a lower bound of $\overline{M}$, then by Proposition 6.1 $\neg q$ is entailed by every lower bound of M, which is to say $M\neg q$ because M is regular. So, to say that p is entailed by every lower bound of $\overline{M}$ is to say that $\neg p$ entails every M-necessary proposition, i.e., it is a lower bound of M. This shows $\overline{M}p$ by Proposition 6.1.

For the last item, assume p is a lower bound of $M \sqcup \overline{M}$. Thus, p entails every conjunction of an M-necessary proposition with an $\overline{M}$-necessary proposition. In particular, it is a lower bound of both M and $\overline{M}$. By Proposition 6.1 it follows that $\overline{M}\neg p$ and $\overline{\overline{M}}\neg p$. Because M is regular, it follows that $M\neg p$. But $M \sqcap \overline{M} = \Box$, so $\neg p = \top$ and so $p = \bot$. □

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[11] Clearly, $\overline{\overline{M \sqcap N}} \leq \overline{\overline{\overline{M} \sqcap \overline{N}}} = \overline{\overline{M}} \sqcap \overline{\overline{N}}$. Conversely, suppose p is entailed by every lower bound of M and by every lower bound of N. Let q be a lower bound of $M \sqcap N$. If Mr and Ns, then $M \sqcap N(r \vee s)$, so $q \leq r \vee s$, in other words $q \wedge \neg s \leq r$. Consequently, $q \wedge \neg s$ is a lower bound of M for any s with Ns. In particular, $q \wedge \neg s \leq p$, which is to say $q \wedge \neg p \leq s$. But s was arbitrary, so we have shown that $q \wedge \neg p$ is a lower bound of N, in particular $q \wedge \neg p \leq p$. But also $q \wedge \neg p \leq \neg p$, so $q \wedge \neg p = \bot$, which is equivalent to $q \leq p$.

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Appendix A. Existence of Regular Non-Boolean Necessities

I prove Theorem 6.6 by giving a model of Classicism that refutes (Boolean Comp) and contains a regular but non-Boolean necessity. We work in the framework of *substitution structures* (Bacon, 2019). A substitution structure is a triple $\mathfrak{A} = (\mathbf{I}, \mathbf{A}, \mathit{Sub})$ where $\mathbf{A}$ is an applicative structure and Sub is an action of the monoid

$\mathbf{I}$ of substitutions on each domain $\mathbf{A}^\sigma$ that commutes with application. Our higher-order language can be recursively interpreted over functional substitution structures with combinators. See Bacon (2019, Def. 8) and Bacon (2024, Chs. 14–15 and Sec. 17.2) for background.

Let S^ω be the set of all infinite binary sequences and let S^n be the set of all binary sequences of length n. Given $s \in S^n$, the *cylinder* of s in S^ω, written $[s]$, is the set of all infinite binary sequences that extend s. Define a topology on S^ω by taking the finite unions of cylinders as the basis. It is well known that the result is a Stone space, hence the dual algebra $\mathsf{Clop}(S^\omega)$ of clopen sets is a Boolean algebra. The clopens are exactly the finite unions of cylinders. In fact, $\mathsf{Clop}(S^\omega)$ is isomorphic to the unique countable atomless Boolean algebra (Givant and Halmos, 2009, Ch. 16, Thm. 10), which is incomplete.

We expand S^ω to a monoid by adding the operation of coordinatewise addition modulo 2, notated $\oplus$. Then $(S^\omega, \oplus)$ is an Abelian group. The identity element is $\mathbf{0} = (0, 0, 0, \ldots)$ and the inverse of any $s \in S^\omega$ is s itself: $s = s^{-1}$.

Now to the applicative structure and action components. We define our domains $\mathbf{A}^\sigma$ and the action *Sub* on elements of $\mathbf{A}^\sigma$ simultaneously by induction on types, writing $\mathbf{fa}$ instead of $App(\mathbf{f}, \mathbf{a})$ and $i_s\mathbf{a}$ instead of $Sub(s, \mathbf{a})$ for readability. Let $\mathbf{A}^e := \{\star\}$ for some arbitrary individual $\star$ and let $i_s\star := \star$ for all $s \in S^\omega$. Let $\mathbf{A}^t := \mathsf{Clop}(S^\omega)$ and define $i_s\mathbf{p}$ by division:

$$i_s\mathbf{p} := \{t \in S^\omega : t \oplus s \in \mathbf{p}\}.$$

Notice that $i_s\mathbf{p}$ is clopen when $\mathbf{p}$ is: i_s maps cylinders to cylinders and preserves finite unions. Moreover, $i_\mathbf{0}\mathbf{p} = \mathbf{p}$ and $i_s(i_t\mathbf{p}) = i_{s\oplus t}\mathbf{p}$, so this is indeed an action as required.

Call a subset $U \subseteq S^\omega$ *finitary* when the following holds: there exists $n \in \omega$ such that for all $s, t \in S^\omega$, if s and t agree on the first n coordinates, then $i_sU = i_tU$. In this case we say that the relevant $n \in \omega$ *pins down* U. We also stipulate that $\star$ is finitary. Observe that the finitary subsets of S^ω are precisely the clopens: each clopen, written as a finite union of cylinders, is pinned down by the maximum length of the sequences generating those cylinders.

Assume inductively that: (i) the domains $\mathbf{A}^\sigma, \mathbf{A}^\tau$ have been defined; (ii) the substitution action on elements of $\mathbf{A}^\sigma, \mathbf{A}^\tau$ has been defined; and (iii) a notion of what it means for an element of $\mathbf{A}^\sigma$ or $\mathbf{A}^\tau$ to be finitary has been defined in such a way that every element of either $\mathbf{A}^\sigma$ or $\mathbf{A}^\tau$ is finitary. We define the domain $\mathbf{A}^{\sigma\to\tau}$ by selecting a subset of the set of all functions $\mathbf{f} : \mathbf{A}^\sigma \to \mathbf{A}^\tau$. Given such an $\mathbf{f}$, for any $s \in S^\omega$ and $\mathbf{a} \in \mathbf{A}^\sigma$ we put

$$(i_s\mathbf{f})\mathbf{a} := i_s(\mathbf{f}(i_{s^{-1}}\mathbf{a})).$$

Since every $s \in S^\omega$ is its own inverse, this equals

$$(i_s\mathbf{f})\mathbf{a} = i_s(\mathbf{f}(i_s\mathbf{a})).$$

We call $\mathbf{f} : \mathbf{A}^\sigma \to \mathbf{A}^\tau$ *finitary* when there exists $n \in \omega$ such that for all $s, t \in S^\omega$, if s and t agree on the first n coordinates, then $i_s\mathbf{f} = i_t\mathbf{f}$. Again, we say that the relevant $n \in \omega$ *pins down* $\mathbf{f}$. We let $\mathbf{A}^{\sigma\to\tau}$ consist of all finitary functions from $\mathbf{A}^\sigma$ to $\mathbf{A}^\tau$. It is routine to verify that $i.$ as just defined is indeed an action on $\mathbf{A}^{\sigma\to\tau}$.

So, $\mathfrak{A} := ((S^\omega, \oplus), (\mathbf{A}^\cdot, App), i.)$ is a substitution structure, with App identified with function application. Clearly, $\mathfrak{A}$ is functional, and has combinators because in the *full* applicative structure based on $\mathbf{A}^t, \mathbf{A}^e$ (where higher-order domains are

given by the full function spaces), combinators exist and are pinned down by 0, hence finitary.

We interpret our higher-order language on $\mathfrak{A}$ by letting Boolean connectives express set-theoretic operations on $\mathbf{A}^t$ and putting

$$[\![\forall_\sigma]\!]^g : \mathbf{f} \mapsto \bigcap\{\mathbf{fa} : \mathbf{a} \in \mathbf{A}^\sigma\}.$$

These interpretations are well defined. The set-theoretic operations as well as the quantifiers are all pinned down by 0. This interpretation is recursively extended to all terms in the usual way. To obtain a model, we define truth by

$$\mathfrak{A} \models \mathbf{p} :\iff \mathbf{0} \in \mathbf{p}.$$

This gives a model of Classicism, because $\mathfrak{A}$ is functional and $\mathfrak{A}$ is *quasi-Fregean*: when for every $s \in S^\omega$ we have $\mathfrak{A} \models i_s\mathbf{p}$ iff $\mathfrak{A} \models i_s\mathbf{q}$, then $\mathbf{p} = \mathbf{q}$.

Now to find a counterexample to (Boolean Comp). Consider the set

$$C := \{\mathbf{0}\} \cup \bigcup\{[0^n10] : n \in \omega\}.$$

Note that C is closed but not open.[12] We use C to define $\mathbf{f} \in \mathbf{A}^{t\to t}$ by

$$\mathbf{fp} := \{s \in S^\omega : i_s(C) \subseteq \mathbf{p}\}.$$

Because $\oplus$ is commutative and every substitution is its own inverse, $\mathbf{f}$ is pinned down by 0 and consequently maps clopens to clopens.

I claim that $\mathbf{f}$ has no greatest lower bound. First, notice that

$$\bigcap\{\mathbf{p} : \mathbf{0} \in \mathbf{fp}\} = C.$$

Now let U be a clopen subset of C. Then U, being compact and not containing 0, intersects only finitely many cylinders of the form $[0^n10]$. But given any such U we can always add another cylinder of the form $[0^m10]$ for some m not already represented in U, and the resulting set will still be clopen and contained in C. So, there is no greatest *clopen* lower bound of the $\mathbf{p}$'s with $\mathbf{0} \in \mathbf{fp}$. Consequently, the present model refutes (Boolean Comp) by taking $\mathbf{f}$ as the counterexample.

Finally, we can show that $\mathbf{f}$ itself is a regular but non-Boolean necessity. Note that $s \in \mathbf{f}(S^\omega)$ iff $i_s(C) \subseteq S^\omega$, which holds for all $s \in S^\omega$. Moreover, it is straightforward to check that $s \in \mathbf{f}(\mathbf{p}\cap\mathbf{q})$ iff $s \in \mathbf{fp}\cap\mathbf{fq}$ holds for any $\mathbf{p}, \mathbf{q} \in \mathbf{A}^t$. Because $\mathfrak{A}$ is functional this implies that $[\![\lambda pq.X(p\wedge q)]\!]^{g[X\mapsto\mathbf{f}]} = [\![\lambda pq.X(p)\wedge X(q)]\!]^{g[X\mapsto\mathbf{f}]}$. So, $\mathbf{f}$ is a necessity. We already know it has no extensional meet, so it is not Boolean. It is, however, regular: for $\mathbf{p}$ to be entailed by every extensional lower bound of $\mathbf{f}$ is for $\mathbf{p}$ to contain every clopen subset of C. The closure of the union of all clopen subsets of C is C itself, so any such $\mathbf{p}$ must contain C. That holds precisely when $\mathbf{0} \in \mathbf{fp}$.

[12]Note that any clopen neighborhood of $\mathbf{0}$ must contain some cylinder of the form $[0^n]$. However, no such cylinder is contained in C: each $[0^n]$ contains some sequence of the form $0^n11\ldots$, which is not in any cylinder of the form $[0^m10]$. This shows that C is not open. To see that it is closed, let U be the union of $\{[0^n10] : n \in \omega\}$ and note that C equals the set of limit points of U. Any $s \in U$ is a limit point of U because U is open and the Cantor space has no isolated points. Moreover, $\mathbf{0}$ is a limit point of U as well: a clopen neighborhood of $\mathbf{0}$ must contain some cylinder of the form $[0^n]$, which in turn contains some sequence of the form $0^n10\ldots$, which is in U. Finally, there are no more limit points of U. If $s \neq \mathbf{0}$ and $s \notin U$, then s must be of the form $0^n11\ldots$. The cylinder $[0^n11]$ is disjoint from U, so $0^n11\ldots$ cannot be a limit point of U.